

Joshua Wu

+1 (404) 971-0536 | jo4467x@gmail.com | <https://joshuawuportfolio.info> | Philadelphia, PA

Education

- University of Pennsylvania** | MSE in Robotics May 2028
- Thesis Option, Certificate of Advanced Scientific Computing
- Georgia Institute of Technology** | BS in Mechanical Engineering GPA 3.94/4.00 Dec 2025
- Concentration in Automation and Robotics, Minor in Computer Science (Computing and Intelligence), Minor in Japanese
- Tokyo Institute of Technology** | Tokyo, Japan March 2024 – Aug 2024
- ACAP Study Abroad program for Mechanical Engineering

Skills

Technical: CAD, Prototyping, Machining, Mechanical and Electrical Systems Testing, and Data Analysis
Tools and Equipment: 3D printing, Oscilloscope, Multimeter, Power Tools, and Soldering Iron
Programming: Python, C/C++, MATLAB, Linux, Java, LaTeX, ROS, Simulink, LabVIEW, and Docker
Software: SolidWorks, GitHub, Arduino, STM32, VS Code, Autodesk Inventor, Onshape, and Autodesk Eagle
Languages: English (native), Chinese-Mandarin (fluent), and Japanese (Intermediate)

Experience

- Aurora– Lidar Signal Processing**, Bozeman, Montana May 2026 – Aug 2026
Hardware Engineering Intern | Linux, Matlab, Python, SLAM
- Created a simulated Signal Processing Pipeline for VnV, conducted before physical testing, saving time and resources
 - Developed a tool to determine vehicle ego velocity based on a point cloud, optimized to be used in real-time
- BRHML Lab Research Assistant**, Atlanta, Georgia Jan 2026 – May 2026
- Assembled a custom signal-interception system using an STM32, motor drivers, and a multiplexer for 3D printer research.
 - Successfully demonstrated vulnerabilities that produced geometric deformations, compromising print strength.
 - Validated signal timing and debugged using oscilloscope and multimeter testing, presenting findings to the research team.
- BAR Lab Research Assistant**, Atlanta, Georgia
- Prototyped and designed Robot End-Effectors to attach to ATI Quick Changers using Solidworks CAD
 - Conducted literature reviews on Non-Destructive Testing Systems
- Southern Spars**, Auckland, New Zealand Jan 2024 – March 2024
Composite Spar Maker | Engineering Drawings, Wet and Pre-preg Carbon Lamination
- Managed and distributed carbon fiber stock across departments increasing productivity in a factory environment.
 - Proofread engineering drawings, successfully preventing errors propagating through manufacturing process.
 - Performed carbon fiber lamination with high efficiency, streamlining production, successfully meeting tight deadlines
- LIDAR Robotics Lab Research Assistant**, Atlanta, Georgia Jan 2023 – Dec 2023
- Prototyped sensor housing on a bipedal robot to protect from harsh environments resulting in uninterrupted operation

Projects

- Self-Stabilizing Airflow Tube Project** | Team Member Aug 2025 – Dec 2025
Feedback control system using Airfoils | Arduino Uno, Firmware
- Integrated sensor feedback using IMU for real-time angle correction
 - Modeled and fabricated the test setup including a suspended tube with adjustable airfoil mounts
- Automatic Card Shuffler Project** | Personal/Capstone Team Lead January 2025 – Present
Autonomous System for shuffling cards | I2C, Electrical Schematic and Documentation Analysis, Leadership
- Independently wrote firmware on STM32IDE.
 - Debugged, troubleshooted, and tested circuit system with Multimeter and Serial Monitoring.
 - Familiarized with tools in the Product Development Lifecycle Including HOQ, BOM, Spec Sheet, Morph Chart
- League of Legends Champion Predictor Project** | Team Lead Aug 2024 – Dec 2024
Devised Algorithm using Logistic Regression, Random Forests, and LSTMs | PyTorch/TensorFlow, scikit-learn, Pandas, Keras
- Processed data collected from over 2000 games via Excel and Python using Data Cleaning and Feature Engineering.
 - Logistic Regression and Random Forest models achieved 92.79% and 97% accuracy respectively

Activities

- Automotive LIDAR** | ROS integration Team Lead August 2025 – Present
Developing a LIDAR system on an RC vehicle | ROS, ROS2, Git, GitLab
- Collaborated with team to migrate projects and code from ROS to ROS2 due to End of Life.
- RoboJackets, RoboWrestling** | Electrical Sub-team Member Aug 2021 – May 2022
Robot-Sumo Competition | Autodesk Eagle, PCB design, Git, GitHub
- Updated PCB for radio board with new routing and component locations, fixing pin access and voltage issues
- Hytech Racing** | Aero-Mechanical Sub-team Member Aug 2021 – Dec 2021
Georgia Tech Formula SAE team | FEA, Teamwork
- Improved Swan-Neck Wing mount using Topology Studies with SolidWorks FEA to reduce material required by 30%